Sample Augmentation With Threshold Estimation for Classification With Hyperspectral Remote Sensed Image

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***Abstract*— Sample augmentation is crucial for improving land-cover classification performance when the samples are limited. However, the traditional sample augmentation approach concentrates on enlarging the quantity of sample via generation and synthetic technique directly, the sample quality is usually neglected. In this article, we propose a novel sample augmentation approach with threshold estimation (SATE) to improve both the quantity and quality of samples for hyperspectral remotely sensed image (HRSI) classification. First, a threshold estimation algorithm (TEA) is proposed to identify high-confidence potential samples from the initial classification map by utilizing the prediction probabilities of different classes. Second, a semi-variational model is employed to detect and correct pseudo-labels in the spatial domain, further enhancing the quality of selected potential samples. Finally, a farthest point sampling (FPS) algorithm optimizes sample distribution in the spectral domain, improving representation for intraclass heterogeneity. Experimental results based on four real HRSIs and compared with eight state-of-the-art few-shot-based methods verify the feasibility and superiority of the proposed SATE approach. The improvement achieved by our proposed approach is about 0.79%–4.31% in terms of the overall accuracy (OA). The code is available at https://github.com/ImgSciGroup/SATE**

***Index Terms*— Few-shot learning, hyperspectral remote sensing imagery, land-cover classification, sample augmentation.**

I. Introduction

THE task of land-cover classification with the hyperspectral remote sensed image (HRSI) classification intends to assign a unique label to each pixel and has been successfully conducted in various applications, such as urban-area monitoring [1], precision agriculture [2], and environmental health evaluation [3]. In the past decades, some representative supervised methods and classifiers, such as support vector machines [4], random forest [5], and supervised neural networks [6], have been successfully applied for HRSI classification. In particular, the quality and quantity of samples are important for a supervised classifier to obtain satisfactory classification performance [7]. Here, the challenges of supervised classification with limited samples are briefly summarized below.

1) HRSI classification with small samples is a challenge due to the intrinsic properties of HRSIs, such as high dimensionality and difficulty in labeling samples [8]. The insufficient samples prevent a classifier from covering the high variability of the data and the complexity introduced by the high dimension of the features. Moreover, the classifier does not properly fit, and its classification performance rapidly degrades. However, labeling many samples for HRSIs is time-consuming and labor-intensive. Therefore, achieving HRSI classification with small initial samples is required and challenging.

2) HRSIs provide rich information but suffer from an inherent interclass imbalance, limiting the practicality of HRSI classification in real-world applications [9]. The HRSI is a record of ground covers on the Earth's surface. Moreover, the HRSI is a typical example of an imbalanced dataset because of the diverse ground covers on the Earth's surface. Therefore, the imbalanced character of HRSIs must be considered when labeling samples for HRSI classification.

3) Mislabeling training samples typically has a negative effect on the performance of HRSI classification [10], [11], [12]. When the referred information for labeling samples is limited, or the land-cover scene is complex for a practitioner, and the HRSIs typically exhibit high intraclass variance, mislabeling is likely to occur. In addition, the error in the location equipment is inevitable [10], [13]. Therefore, mislabeled samples may be present in the sample set, and how to correct and revise them is a significant challenge.

In recent years, various HRSI classification methods have been proposed to improve the classification performance of HRSIs when samples are limited. Here, these methods can be divided into four aspects: sample augmentation [14], deep learning-based methods [8], sample imbalance, and detection

Received 29 December 2024; revised 27 April 2025; accepted 20 May 2025. Date of publication 26 May 2025; date of current version 12 June 2025. This work was supported in part by the National Natural Science Foundations of China under Grant 42104012, Grant 42474021, and Grant 42271385; in part by the Natural Science Basic Research Programs of Shaanxi under Grant 2023-YBGY-236 and Grant 2023-JC-YB-572; in part by the Key Laboratory of Urban Land Resources Monitoring and Simulation, Ministry of Natural Resources, under Grant KF-2022-07-010; and in part by Shenzhen Science and Technology Program under Grant JCYJ20220531101812028 and Grant 20231121094609001. *(Corresponding author: Xiaoqiong Qin.)*

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Digital Object Identifier 10.1109/TGRS.2025.3573473

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and correction of pseudo-samples. The details of these methods are reviewed below.

1) Sample augmentation is an intuitive and straightforward way to improve the performance of HRSI classification when the training samples are limited [15]. In particular, the pseudo-label technique is the most common sample augmentation strategy, which fully uses massive unlabeled samples [16], such as a semi-supervised framework that selects unlabeled samples whose predicted probabilities are higher than a predefined threshold for all classes [17], a sample query based on active learning algorithms [18], [19], and a mining sample based on distance metrics and diversity criterion [20], [21]. However, the above-mentioned methods adopt a predefined threshold to select unlabeled samples, but they do not consider different learning statuses and learning difficulties of diverse classes [16], [22]. In addition, another sample augmentation increases the samples by virtual sample generation, including transformation operations [23], mixup [24], and occlusion [25]. However, the main limitation of the above-mentioned methods is that the performance largely depends on the quality of generated samples, which is hard to control or standardize [26]. In addition, the generative adversarial network (GAN) is an effective way to generate virtual samples by training two neural networks to compete with each other, thereby generating more realistic new samples from a given training set [27]. For instance, the 3-dimensional GAN (3D-GAN) directly performs the HRSI classification task using original and generated fake samples [28]. The adaptive dropblock-enhanced GAN (ADGAN) combines a dropblock with a GAN to generate fake samples that are spatially well-distributed [29]. Although various studies indicate that the GAN can create new samples for enhancing classification performance, these methods only consider the importance of the sample's quantity for HRSI classification, and the negative impact of the imbalance for intraclasses in the sample generation process is ignored completely. Moreover, the generated samples may not represent the distribution of the real original samples [30].

2) Besides sample augmentation methods, numerous deep learning-based methods have been proposed for HRSI classification with small samples. For example, the few-shot learning network aims at learning the differences among the samples instead of directly learning the sample's features [31], [32]. Several studies have introduced the few-shot learning into HRSI classification with small samples, such as the deep few-shot learning (DFSL) [33], the spectral–spatial Siamese network (S3-Net) [34], few-shot graph convolution [35], the multitask deep learning method (MDL-Net) [36], the learning compact and discriminative stacked autoencoder [37], [38], and the multistage relation network with dual-metrics (DM-MRNs) [39], to achieve this objective. In addition, meta-learning frameworks based on meta-knowledge [40] and holistic prototype activation strategies [41] provide novel insights for addressing the scarcity of labeled samples in HRSI classification. In contrast to sample augmentation, few-shot learning improves the HRSI classification performance with limited samples via optimizing the structure of the neural network and training strategy [42].

3) Numerous approaches also concentrate on the imbalance problem in the community of HRSI classification [43], [44]. Mani and Zhang [45] selected some representative samples from the majority classes based on k-nearest neighbors undersampling algorithms. The synthetic underrepresented groups over-sampling technique (SMOTE) [46], adaptive synthetic sampling [47], and borderline SMOTE [48] are developed to generate new samples to populate the underrepresented groups classes. The generated samples are linear combinations of original samples that may not express class distribution. Recent work focuses on the GAN generation of underrepresented groups class samples, such as a deep generative spectral–spatial classifier [49]. Generative adversarial underrepresented groups oversampling (3D-HyperGAMO) [50] has shown great performance in imbalanced HRSI classification problems. However, even though the data-level balance is achieved, the model may result in an imbalanced classification without regarding different classes of learning status and difficulties [22], [51], [52].

4) The detection and correction of pseudo-samples are important for improving classification performance with HRSIs, including pseudo-sample tolerant algorithms [12] and pseudo-sample detection methods [53]. Pseudo-sample labeling is an important method in semi-supervised learning [11], [13]. Predicting and labeling unlabeled samples and combining them with the initially labeled set significantly improves model performance. Because the model prediction is not accurate, it is necessary to detect mislabeled samples. Existing studies on a pseudo-sample-tolerant approach typically focus on modifying class probabilities or correcting loss functions [10]. By contrast, the pseudo-sample correction approach aims at revising the label of the noisy samples [11]. In particular, numerous pseudo-sample correction methods, including the density peak-based noisy label detection [11], the pseudo-sample detection based on spectral–spatial graphs [11], [12], the hierarchical structure noisy label detection [10], and the pseudo-sample detection based on superpixel-to-pixel weighting distance [13], have been successfully applied for HRSI classification with noisy labels. The above-mentioned approaches all need to first detect the label noise samples and then correct them, and thus, their performances are inevitably affected by the detection algorithm. In addition, the detection and removal strategy will largely reduce the number of training samples, especially when the noise level is high [12].

Based on the above literature review, it can be seen that many sample augmentation approaches generate or synthesize samples directly [49], and these studies demonstrate that improving the number of samples via the sample augmentation approach can enhance the classification performance when samples are limited and small. Despite generating and synthesizing samples that can augment the sample's quantity, the sample's quality is usually neglected. The distribution of each image is unique, the targets are usually imbalanced in an image scene [51]. Moreover, interclass heterogeneity exists in remote sensing images and should be considered when labeling samples for classification. Therefore, an approach named sample augmentation approach with threshold estimation (SATE) is proposed to enhance the sample's quality and quantity for

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improving land-cover classification performance with remote sensing images. The difference between the proposed approach and the existing sample augmentation lies in the following two aspects.

1) Lots of existing methods select the pixels with high-confidence labels with a fixed threshold for all the classes. The operation is dogmatic because the classes in images are usually imbalanced and they are different in separability. Therefore, the threshold estimation technique is required to select the pixel with the high-confidence label for each class.
2) Most existing methods focus solely on increasing the number of samples for each class. In our proposed approach, we not only address the quantity of samples per class but also prioritize sample quality.

This ensures coverage of both interclass and intraclass heterogeneity. The major contributions of this work can be summarized as follows.

1) A threshold estimation algorithm (TEA) is proposed to explore the pixels with high confidence to expand the initial samples. The TEA is first proposed to extract the potential samples from a classified map. Moreover, the TEA has an advantage in adaptively adjusting the threshold for selecting the classified pixels with high confidence from each class.

2) Pseudo-label detection and correction (PLDC) is proposed to reduce the negative influence of the noise samples explored by the TEA. In the proposed PLDC, the semi-variogram is first adopted to detect and correct the samples with pseudo-labels. To the best of our knowledge, the effectiveness and feasibility of the semi-variogram for PLDC are first verified in the community of classification with HRSIs.

3) The sample's spatial distribution is first optimized by the farthest point sampling (FPS) algorithm [54]. Considering the phenomenon of intraclass heterogeneity in remotely sensed images, the idea of FPS inspires us to further improve the representative ability of samples.

This article is organized as follows. The SATE is introduced in Section II. The comparative and ablation experiments are presented in Section III. The discussions and analysis are presented in Section IV. The conclusions are drawn in Section V.

II. Method

Motivated by the idea of utilizing intraclass heterogeneity and prediction probability mapping from a classifier, an SATE approach is proposed to enhance the classification performance with HRSIs when initial samples are very small. The details of the proposed SATE approach are introduced below.

A. Overview

First, we present an overview of the proposed approach to provide an entire understanding. All the datasets used in the experiments are publicly accessible datasets for evaluating the classification performance, and the preparations and processing for these datasets have been done before releasing the dataset. According to the ground truth of the dataset, a small number of samples of each category are manually selected to form the initial small-sample set. The supervised classifier called the Siamese network classifier is utilized to acquire a class-prediction probability map (Fig. 1). Thereafter, the TEA is proposed to explore the candidate samples from the probability map. Finally, two new strategies, namely PLDC with a semi-variogram and the sample's spatial distribution optimization with the FPS algorithm, are proposed to further improve the sample's quality from the perspectives of the sample's heterogeneity. The details of each part will be elaborated below.

B. Sample Augmentation With the TEA

The separability of the referred classes in an image scene varies, and exploring samples with a hard threshold of each class is unreasonable [55], [56]. Accordingly, a TEA is required to explore the potential samples from an initial classified probability map. For clarity, an initial small-sample set $S_t = \{S_t^{c_1}, S_t^{c_2}, S_t^{c_3}, \ldots, S_t^{c_n} \mid t = 0, 1, 2, \ldots, \}$ is manually prepared when $t = 0$, where n denotes the total number of classes and $S_t^{c_n}$ is a sample subset for the c_nth class in the tth iteration. In the proposed TEA, an initial threshold set $\tau_t = \{\tau_t^{c_1}, \tau_t^{c_2}, \tau_t^{c_3}, \ldots, \tau_t^{c_n}\}$ is defined to select pixels with a high-confidence label. For each class, pixels whose predicted probability exceeds the corresponding threshold are counted. For example, if a pixel is predicted to be labeled c_1, and its predicted probability is larger than $\tau_t^{c_1}$, the pixel will be counted. When the whole classified image is processed in this way, the number of pixels with high-confidence prediction probability of each class can be normalized as $\beta_t = \{\beta_t^{c_1}, \beta_t^{c_2}, \beta_t^{c_3}, \ldots, \beta_t^{c_n}\}$. Based on this, τ_{t+1} is updated by the following equation:

$$\tau_{t+1}^{c_k} = \varphi\left(\beta_t^{c_k}\right) \cdot \tau_t^{c_k}. \tag{1}$$

To ensure each class acquires sufficient learnable samples, thresholds for classes with smaller β_t are lowered in subsequent iterations, while those for classes with larger β_t remain unchanged. To calculate the threshold of each class for the next iteration, we design $\varphi(\cdot)$, which is a decreasing convex mapping function and can be calculated by the following equation:

$$\varphi\left(\beta_t^{c_k}\right) = -\alpha\left(\beta_t^{c_k} - 1\right)^2 + 1 \tag{2}$$

where $\alpha \mid \alpha \in (0, 1)$ is a customized parameter. A larger α means a smaller threshold in the next iteration for the class c_k. By contrast, a smaller α will produce a larger threshold. Therefore, the threshold set τ_t will be updated in each iteration, and the iterative progress will be determined by the following equation:

$$\tau_t^{c_k} < \frac{1}{n}. \tag{3}$$

In the above discussion, the labels of the sample may be unreliable when (3) is satisfied. Accordingly, the threshold update operation of this class will be terminated, and the candidate sample set S_t' will be selected from the classified map in the last iteration when all the classes satisfied (3).

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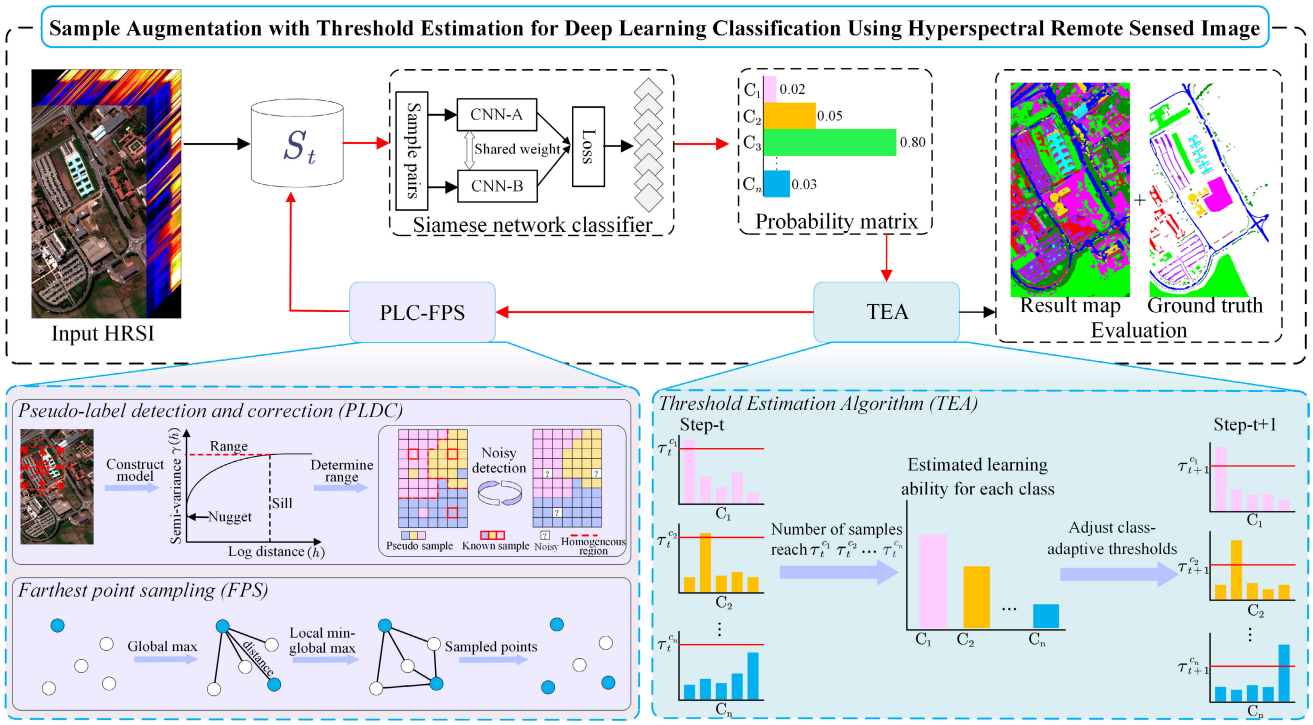


Fig. 1. Flowchart of the SATE for deep learning classification with HRSIs.

C. PLDC With a Semi-Variogram

Some pixels may be misclassified in a classification map. Accordingly, noisy labels may exist in the candidate sample set S_t'. To further improve the augmented sample's quality in S_t'. Here, we assume that the pixels that make up a geographic object are spatially continuous and spectrally similar, and the spatial variation pattern of these pixels can be modeled by the semi-variogram function. Accordingly, a PLDC approach based on a semi-variogram is promoted to improve the sample's quality from the spatial domain. The semi-variogram is a function that describes the degree of spectral variance in a spatial domain. In the field of geostatistics, it is used to analyze the spatial variation characteristics in a geographic target [57]. Unlike the numerous applications of semi-variograms, a significant motivation for the proposed PLDC based on semi-variograms lies in describing the diversification of intraclass samples to cover the intraclass heterogeneity in the spatial domain. The details are presented below.

The position of the known samples for class c_k is defined as $S_t^{c_k'} = \{j_1, j_2, j_3, \ldots, j_m\}$, m is the total number of samples in $S_t^{c_k'}$, and semi-variance $\gamma(h)$ is represented by the following equation:

$$\gamma(h) = \frac{1}{2n(h)} \sum_{h=1}^{n(h)} \left[x(j) - x(j+h)\right]^2 \tag{4}$$

where h is the spatial distance sampling interval in the spatial domain; $x(j)$ and $x(j+h)$ denote the spectral values of the samples at positions j and $j+h$, respectively; and $n(h)$ is the number of pairs of pixels $x(j)$ and $x(j+h)$ in the HRSI. The SATE will be applied in eight directions to remove the noisy labels, including $\{0°, 45°, 90°, \ldots, 315°\}$, to account for the distribution of samples in their corresponding spatial domain. In each direction (Fig. 2), the samples outside the red area will be defined as the pseudo-samples and excluded from the candidate sample set S_t'.

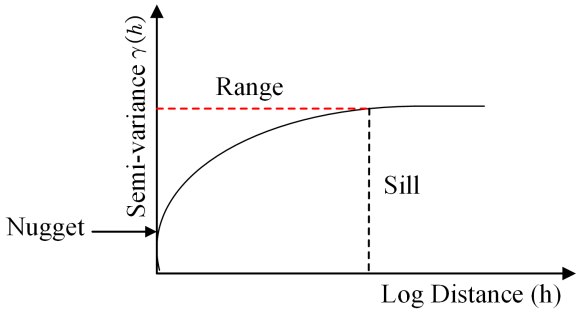


Fig. 2. Semantic demonstration of the proposed PLDC with a semi-variogram.

D. Sample's Spectral Distribution Optimization With an FPS Algorithm

The above statement implies that pseudo-samples can be removed through the constraint of semi-variograms in the spatial domain. However, HRSIs often exhibit spectral redundancy in local regions. To further reduce redundancy and improve the reliability of pseudo-samples, some potential samples can be filtered through a multivariate fuzzy k-means optimal sampling algorithm [58]. Accordingly, an FPS strategy [54] is adopted here for sample selection in our proposed approach in set S_t' to consider the spectral heterogeneity and minimize redundancy. FPS is first proposed for the sampling and processing of point

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cloud data. This mechanism has several advantages, such as maintaining intraclass sample uniformity, efficiently covering intraclass spectral variability, and providing an approximation of ground objects with fewer samples. The details of the FPS applied here for sample optimization are as follows.

1) Given a candidate sample set $S_t^{c_k'}$, the total number of samples in $S_t^{c_k'}$ is denoted as m and μ is defined as the sampling rate.
2) Pixel $v_0' \in S_t^{c_k'}$ is randomly selected, and a sampling point set $S_t^{c_k''} = \{v_0'\}$ is obtained.
3) An array L with m-dimension is constructed, which stores the spectral Euclidean distance from all samples in $S_t^{c_k'}$ to v_0'. The sample corresponding to the maximum value in L is selected as v_1', and $S_t^{c_k''} = \{v_0', v_1'\}$ is updated.
4) The distance from all samples in $S_t^{c_k'}$ to v_1' is calculated. In point v_i', if $d(v_i', v_1')$ is less than $L(i)$, then $L(i) = d(v_i', v_1')$ was updated. The sample corresponding to the maximum value in L is selected as v_2', and $S_t^{c_k''} = \{v_0', v_1', v_2'\}$ is updated.
5) Step 4) was repeated. FPS stops sampling until the sampling rate reaches μ, and the sampling point set $S_t^{c_k''}$ is obtained. $S_t'' = \{S_t^{c_1''}, S_t^{c_2''}, S_t^{c_3''}, \ldots, S_t^{c_n''}\}$ denotes the expanded sample set for this round of iteration, and S_{t+1} is constructed by appending S_t'' to the initial sample set S_t.

The SATE minimizes reliance on manual annotation while ensuring robust performance. By systematically enhancing high-quality samples and improving data diversity, support the SATE to achieve robust classification performance in various scenarios. However, the SATE still faces challenges that point to future research directions. One limitation is that although the SATE greatly reduces the reliance on manual annotation, it still requires an initial set of high-quality annotations. Addressing these challenges through unsupervised or weakly supervised approaches can further improve the scalability and applicability of the SATE. Moreover, since the number of classes and coverage of HRSI data are significantly different from each other, in practice, the optimal configuration of parameters α and μ needs to be determined by trial and error. This will incur additional time expenditure.

III. Experiment

In this section, some experiments are designed to verify the feasibility of the proposed approach with some real datasets. Eight state-of-the-art methods are used for comparison with the proposed approach to demonstrate its advantages in improving classification performance. The details are presented as follows.

A. Datasets Description

The proposed SATE and state-of-the-art methods are verified and compared on four benchmark datasets, namely Pavia University (Dataset-1), Salinas (Dataset-2), WHU-Hi-HanChuan (Dataset-3), and Houston University (Dataset-4). The detailed description of each dataset is summarized in Table I, and the false-color image and the corresponding ground reference map are illustrated in Fig. 3. These datasets are acquired by using different sensors, including the ROSIS sensor, the AVIRIS sensor, the Headwall Nano-Hyperspec sensor, and the compact airborne spectrographic sensor. The spatial resolution included 0.1, 0.32, 2.5, and 3.7 m/pixel. Therefore, achieving satisfactory classification maps with limited samples for the datasets is a challenging task.

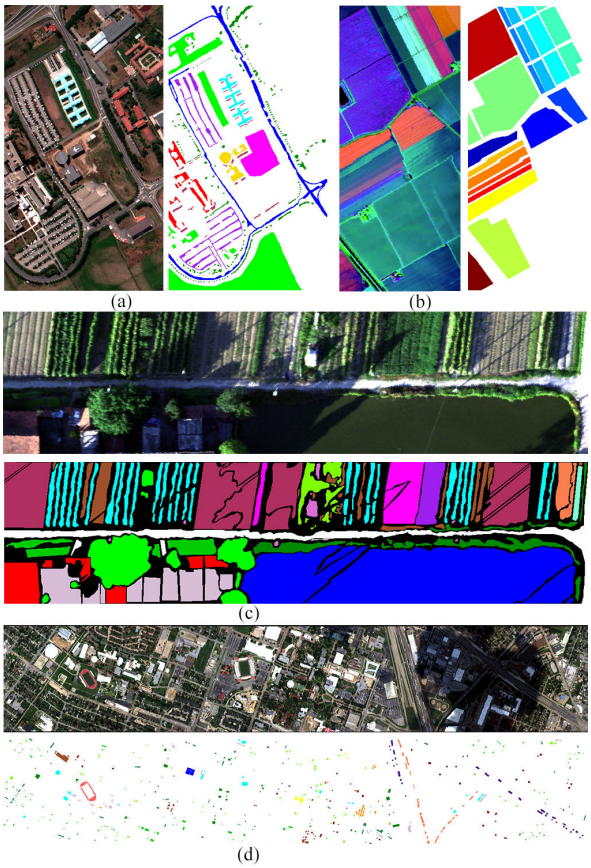


Fig. 3. Experimental dataset shown with three-band false-color and the corresponding ground reference map. (a) Dataset-1. (b) Dataset-2. (c) Dataset-3. (d) Dataset-4.

B. Experimental Settings and Evaluation Metrics

In the experiments, several few-shot learning methods, which are widely used for land-cover classification when the training samples are limited, are selected for comparison. These methods include a 3-D convolutional neural network (3D-CNN) [59], DFSL [33], S3-Net [34], MDL-Net [36], and DM-MRN [39]. In addition, three representative methods based on the GAN for HRSI classification, 3D-GAN [28], 3D-HyperGAMO [50], and ADGAN [29] are compared with the proposed approach to verify the feasibility and advantages of the proposed approach. All comparison methods are retrained from scratch without using pretrained models. The parameter configurations of the comparison method are kept the same with the corresponding publications. Each approach and parameter configuration are briefly described as follows.

1) *3D-GAN [28]:* 3D-GAN introduces a GAN based on two 3D-CNNs into HRSI classification for the first time.

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TABLE I
DETAILED DESCRIPTION FOR EACH DATASET

Dataset	Location	Platform and Sensor	Size (pixel)	Resolution (m/pixel)	Total classes
Pavia University (Dataset-1)	Pavia University, Italy	Aerial planes and ROSIS sensor	$610 \times 340 \times 103$	0.32	9
Salinas (Dataset-2)	California, USA	AVIRIS sensor	$512 \times 217 \times 224$	3.70	16
WHU-Hi-HanChuan (Dataset-3)	Hubei, China	Headwall Nano-Hyperspec imaging sensor	$1217 \times 303 \times 274$	0.10	16
Houston University (Dataset-4)	Houston University, USA	Compact airborne spectrographic imager	$601 \times 2384 \times 48$	2.50	20

A 3D-CNN is designed to discriminate the inputs, and another 3D-CNN is used to generate samples. This type of adversarial training improves the generalization capability of the discriminative 3D-GAN, which is important when the training samples are limited. The parameters of the 3D-GAN are configured as epoch = 500, patch size = 9 × 9, and RBF−SVM as the classifier.

2) *3D-HyperGAMO [50]:* The 3D-HyperGAMO is an effective underrepresented groups class over-sampling technique for handling the class imbalance in HRSI classification. This framework consists of a 3D-generator network that uses 3-D patches and a middle-level representation produced by a conditional feature mapping unit to generate new samples. The number of samples to be generated is decided by the difference between the majority classes and underrepresented groups classes. The parameters of 3D-HyperGAMO are configured as epoch = 8000, patch size = 9 × 9, and learning rate = 0.0002.
3) *ADGAN [29]:* ADGAN can effectively alleviate the problem of interclass imbalance in HRSI classification by adjusting the discriminator to a single output to avoid self-contradiction. Furthermore, the AdapDrop algorithm is designed to effectively handle ground objects of diverse shapes during the sample generation. The parameters of the ADGAN are configured as patch size = 27 × 27 and $K = 40$.
4) *3D-CNN [59]:* A classic 3D-CNN architecture for HRSI classification simultaneously processes the spatial and spectral components using real 3-D convolutions, resulting in the improved use of the limited samples available with fewer training parameters. The parameters of the 3D-CNN are configured as epoch = 200, patch size = 9 × 9, and learning rate = 0.001.
5) *DFSL [33]:* The key insight is that the DFSL method trains a network (a deep residual 3-D CNN) to learn a metric space, where samples from the same class are close, and those from different classes are separated by distance. In such a space, classification can be conducted by using a simple classifier (e.g., K-nearest neighbor and support vector machines). The parameters of the 3D-CNN are configured as epoch = 500, patch size = 9 × 9, learning rate = 0.001, NC = 20, and NQ = 19.
6) *S3-Net [34]:* S3-Net is a lightweight few-shot learning model based on a Siamese structure for HRSI classification. The contribution of the S3-Net lies in the design of a lightweight spatial–spectral feature extractor, a differentiated input strategy for the Siamese network framework, and a weighted contrastive loss, all of which contribute to enhancing the discrimination power of the S3-Net. The parameters of the 3D-CNN are configured as epoch = 50, patch size = 9 × 9, and learning rate = 0.01.
7) *MDL-Net [36]:* MDL-Net is first introduced to identify unknown classes of HRSIs. Given the lack of labels, the MDL-Net defines a threshold based on extreme value theory to distinguish between unknown and known classes. The MDL-Net also proposes two strategies for few-shot scenarios. The parameters of the MDL-Net are configured as epoch = 200, learning rate = 0.1, reconstruction loss weights = 0.5, and unknown probability threshold = 0.5.
8) *DM-MRN [39]:* DM-MRN constructs a multistage network with two metrics. An adaptive weighting strategy is devised to merge multilevel features, addressing the challenge where a single metric may not effectively represent the class distribution with limited labeled samples. The parameters of the MDL-Net are configured as epoch = 1000, patch size = 7 × 7, learning rate = 0.01, and $K = 5$.

The parameters of the proposed approach are settled below for each dataset: to ensure the fairness of the experiment, ten samples of each category are manually labeled as initial samples for each method. $\tau_0 = 0.9$ is fixed for all the datasets; $\alpha = 0.2$ for Dataset-1 and Dataset-2 and $\alpha = 0.3$ for Dataset-3 and Dataset-4; $\mu = 0.3$ for Dataset-2 and Dataset-3 and $\mu = 0.4$ for Dataset-1 and Dataset-4. The Siamese network is constructed with two three-layer CNNs, adopted cross-entropy loss, and the patch size is fixed at 9×9. An Adam optimizer with a 10^{-5} learning rate is used to train the SATE under 100 iterations. All the experiments are implemented in Python 3.7 and PyTorch 1.13 with a GeForce NVIDIA 1080Ti.

Four widely used indicators, including overall accuracy (OA) [60], average accuracy (AA) [61], kappa coefficient (Ka) [62], and $F1$-score [61] are adopted for quantitative comparisons. In addition, the standard deviation of the user accuracy (SDUA) [63] is adopted to evaluate the accuracy-balanced ability for each class. The detailed descriptions of the evaluation metrics are summarized in Table II.

C. Experimental Results

The quotative comparisons are summarized in Tables III–VI. Certain phenomena can be observed in the quantitative comparisons as follows.

1) The proposed SATE achieves improved land-cover classification performance when the training samples are limited. The proposed SATE approach achieves 3.39%–12.53% improvement in terms of OA for the four datasets compared to baseline networks, such as the 3D-CNN and 3D-GAN, which only use a few samples

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TABLE II
BRIEF SUMMARY OF THE APPLIED EVALUATION METRICS

Evaluation Indicators	Formula	Explanation
Overall Accuracy (OA)	$OA = \frac{TP+TN}{TP+TN+FP+FN}$	OA is the correctly detected pixels between the total pixels of the ground map.
Average Accuracy (AA)	AA = $(\frac{TP}{TP+FP} + \frac{TN}{TN+FN})/2$	AA is the mean of correctly detected changed and correctly unchanged ratios.
Kappa Coefficient (Ka)	$P_0 = \frac{TP+TN}{TP+TN+FP+FN}$ $P_e = \frac{(TP+FP)\times(TP+FN)+(FN+TN)\times(FP+TN)}{(TP+TN+FP+FN)^2}$ $Ka = \frac{P_0-P_e}{1-P_e}$	Ka reflects the reliability of the detection map by measuring inter-rater reliability for changed and unchanged classes.
F1-score	$F1 = 2 \times \frac{Pr \times Re}{Pr+Re}$	F1-score is the harmonic mean of precision and recall.
Standard Deviation User's Accuracy (SDUA)	$SDUA = \sqrt{\frac{1}{T}\sum_{i=1}^{i=T}\left(UA_i - \overline{UA}\right)^2}$	SDUA is the standard deviation of user's accuracy.

TABLE III
QUANTITATIVE COMPARISON BETWEEN THE SELECTED METHODS AND THE PROPOSED APPROACH FOR DATASET-1 (# DENOTES THE TOTAL NUMBER OF ITERATIONS OF THE PROPOSED APPROACH). THE BEST RESULTS ARE PRESENTED IN BOLD FONT

Method	OA (%)	AA (%)	Ka	SDUA	F1-score (%)
3D-GAN [28]	82.24	84.44	0.756	11.92	74.43
3D-HyperGAMO [50]	90.86	87.59	0.878	9.08	85.64
ADGAN [29]	88.96	82.64	0.856	15.41	83.78
3D-CNN [59]	80.64	77.70	0.750	11.67	79.55
DFSL [33]	85.41	88.34	0.810	13.12	85.67
S3-Net [34]	89.87	87.03	0.866	9.57	88.17
MDL-Net [36]	90.95	87.33	0.881	6.70	89.10
DM-MRN [39]	84.98	77.27	0.803	14.80	78.22
Proposed (#9)	**93.17**	**91.71**	**0.927**	**5.30**	**92.49**

TABLE IV
QUANTITATIVE COMPARISON BETWEEN THE SELECTED METHODS AND THE PROPOSED APPROACH FOR DATASET-2 (# DENOTES THE TOTAL NUMBER OF ITERATIONS OF THE PROPOSED APPROACH). THE BEST RESULTS ARE PRESENTED IN BOLD FONT

Method	OA (%)	AA (%)	Ka	SDUA	F1-score (%)
3D-GAN [28]	89.55	94.70	0.883	10.55	92.61
3D-HyperGAMO [50]	91.00	93.33	0.900	8.16	93.65
ADGAN [29]	90.95	94.15	0.899	12.97	92.60
3D-CNN [59]	86.92	93.83	0.855	12.86	93.49
DFSL [33]	88.81	92.73	0.875	8.79	90.60
S3-Net [34]	**93.85**	95.38	**0.931**	5.27	96.28
MDL-Net [36]	92.82	94.84	0.920	7.72	95.05
DM-MRN [39]	88.46	88.01	0.871	10.92	88.78
Proposed (#10)	93.48	**96.19**	0.927	**5.01**	**96.54**

TABLE V
QUANTITATIVE COMPARISON BETWEEN THE SELECTED METHODS AND THE PROPOSED APPROACH FOR DATASET-3 (# DENOTES THE TOTAL NUMBER OF ITERATIONS OF THE PROPOSED APPROACH). THE BEST RESULTS ARE PRESENTED IN BOLD FONT

Method	OA (%)	AA (%)	Ka	SDUA	F1-score (%)
3D-GAN [28]	75.88	66.48	0.718	21.6	59.19
3D-HyperGAMO [50]	82.01	67.04	0.790	21.19	66.88
ADGAN [29]	79.11	69.67	0.755	21.51	63.01
3D-CNN [59]	73.29	59.88	0.693	20.45	61.84
DFSL [33]	80.52	67.20	0.774	21.17	70.07
S3-Net [34]	83.53	72.67	0.809	19.82	75.61
MDL-Net [36]	84.76	73.62	0.822	19.87	76.06
DM-MRN [39]	80.01	68.39	0.769	21.96	70.68
Proposed (#12)	**85.57**	**74.79**	**0.834**	**19.12**	**76.44**

TABLE VI
QUANTITATIVE COMPARISON BETWEEN THE SELECTED METHODS AND THE PROPOSED APPROACH FOR DATASET-4 (# DENOTES THE TOTAL NUMBER OF ITERATIONS OF THE PROPOSED APPROACH). THE BEST RESULTS ARE PRESENTED IN BOLD FONT

Method	OA (%)	AA (%)	Ka	SDUA	F1-score (%)
3D-GAN [28]	78.55	71.68	0.768	14.21	71.45
3D-HyperGAMO [50]	84.61	85.56	0.834	9.21	85.63
ADGAN [29]	82.88	86.8	0.815	14.03	85.47
3D-CNN [59]	34.02	33.96	0.290	15.37	32.03
DFSL [33]	83.05	86.13	0.816	13.31	83.74
S3-Net [34]	86.21	87.81	0.851	11.74	88.05
MDL-Net [36]	86.82	87.66	0.857	11.26	87.48
DM-MRN [39]	67.13	67.19	0.646	12.74	66.93
Proposed (#12)	**88.22**	**90.13**	**0.872**	**9.01**	**89.61**

for training. The results indicate that the proposed SATE approach improves the land-cover classification accuracies when the training samples are limited. The SATE achieves improvement rates of approximately 2.22%–8.19% and 0.81%–5.56% for Datasets-1 and Datasets-3 in terms of OA compared to the few-shot methods, such as the DFSL, S3-Net, MDL-Net, and DM-MRN. Furthermore, the performance of the proposed SATE approach demonstrates improved balanced ability for user accuracies. For example, the SATE outperforms the underrepresented groups oversampling strategy in 3D-HyperGAMO, resulting in improved SDUA scores across the four datasets by 3.78%, 3.15%, 2.07%, and 0.20%. All the quantitative comparisons are summarized in Tables III–VI to provide clear comparisons. The SATE in Table IV is inferior to the S3-Net in terms of OA and Kappa evaluation metrics, because the S3-Net focuses on learning differences among data, instead of a direct projection between data and labels. It aims to decrease feature distance within the same class and increase feature distance between different classes. In other words, the S3-Net can learn deep features and augment samples. Dataset-2 covers 16 crops, and the interclass confusion is large. Compared with using single sample augmentation (our proposed approach), the S3-Net has advantages in improving classification accuracies in terms of OA and Ka. However, it is worth noting that the proposed approach which concentrates singly on sample augmentation also achieves competitive accuracies (OA = 93.48% and Ka = 0.927%). Moreover, the SATE has a slight improvement in terms of SDUA compared to that of the S3-Net, which means that the SATE has an advantage in balancing the user accuracy for each class.

2) The visual performance further supports the conclusion based on the quantitative comparisons. Figs. 4–7 demonstrate that the proposed SATE obtains the classification maps with minimal noise, indicating superior visual performance. For example, in the classification

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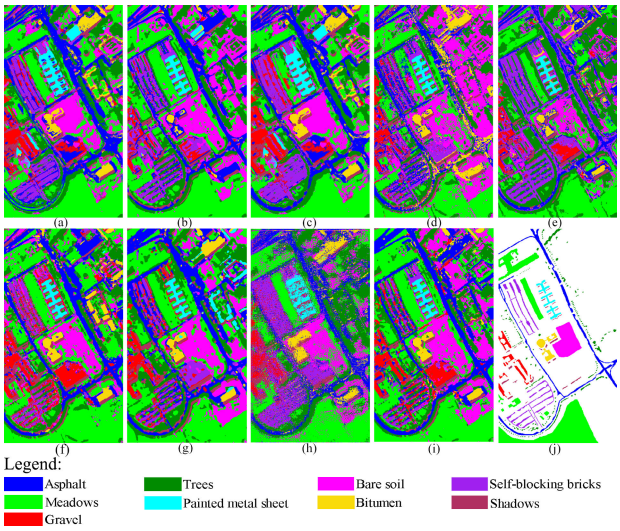


Fig. 4. Classification maps obtained with different traditional methods for Dataset-1. (a) 3D-GAN [28], (b) 3D-HyperGAMO [50], (c) ADGAN [29], (d) 3D-CNN [59], (e) DFSL [33], (f) S3-Net [34], (g) MDL-Net [36], (h) DM-MRN [39], (i) proposed approach, and (j) ground truth.

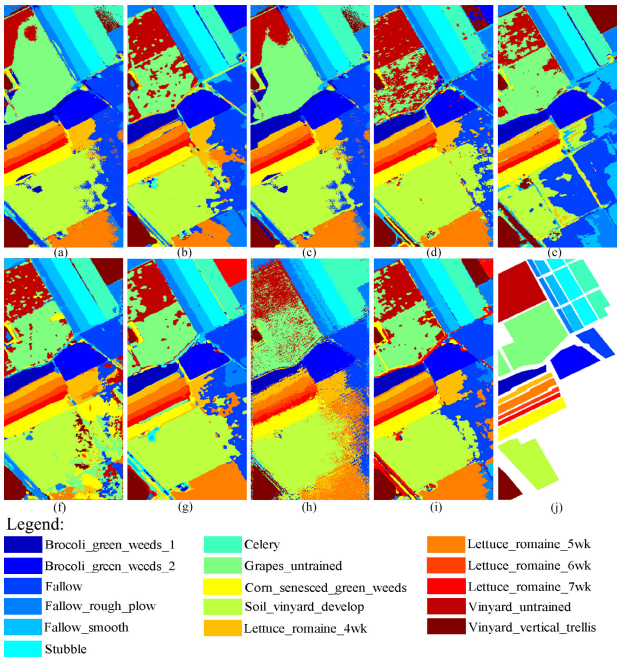


Fig. 5. Classification maps obtained with different traditional methods for Dataset-2. (a) 3D-GAN [28], (b) 3D-HyperGAMO [50], (c) ADGAN [29], (d) 3D-CNN [59], (e) DFSL [33], (f) S3-Net [34], (g) MDL-Net [36], (h) DM-MRN [39], (i) proposed approach, and (j) ground truth.

map obtained by the proposed SATE, the representations of the meadows and bare soil in Dataset-1, grape and vineyard untrained in Dataset-2, and soybean and water in Dataset-3 closely resemble the ground-truth maps. Moreover, the result achieved by the SATE indicates its effectiveness in handling complex scenes (Fig. 7).

D. Ablation Experiments

Ablation experiments are conducted by adding each proposed module to the basic framework to verify the feasibility of the modules. The proposed modules are named TEA, PLDC, and FPS. The ablation experiment results in Table VII demonstrate that each module can effectively improve the classification accuracy when combined with the basic framework. For example, the basic framework achieves OA = 85.43%, 87.71%, 79.04%, and 77.80% for Dataset-1, Dataset-2, Dataset-3, and Dataset-4, respectively. When the basic framework is coupled with the TEA, the OA is greatly improved by 4.55%, 4.07%, 4.36%, and 6.15%, respectively. Benefiting from the noise sample detection of PLDC, the classification accuracy of Dataset-1 and Dataset-2 reaches more than 90% in terms of OA. When the basic framework is embedded with the three modules (TEA, PLDC, and FPS modules), the accuracy is higher than those attained with some modules and without any modules. The observation based on the ablation experiments indicates the effectiveness and feasibility of the proposed modules in improving classification performance with remotely sensed images when training samples are limited.

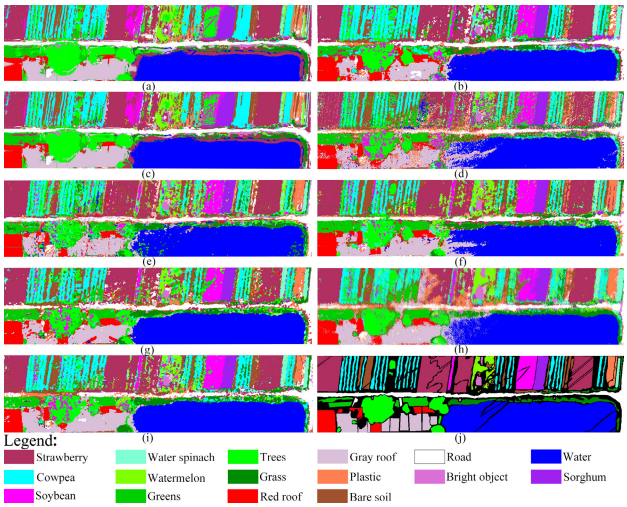


Fig. 6. Classification maps obtained with different traditional methods for Dataset-3. (a) 3D-GAN [28], (b) 3D-HyperGAMO [50], (c) ADGAN [29], (d) 3D-CNN [59], (e) DFSL [33], (f) S3-Net [34], (g) MDL-Net [36], (h) DM-MRN [39], (i) proposed approach, and (j) ground truth.

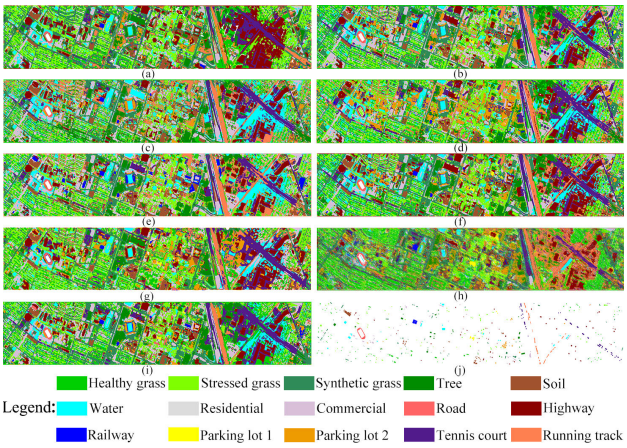


Fig. 7. Classification maps obtained with different traditional methods for Dataset-4. (a) 3D-GAN [28], (b) 3D-HyperGAMO [50], (c) ADGAN [29], (d) 3D-CNN [59], (e) DFSL [33], (f) S3-Net [34], (g) MDL-Net [36], (h) DM-MRN [39], (i) proposed approach, and (j) ground truth.

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TABLE VII
ABLATION EXPERIMENT ON DIFFERENT DATASETS

Dataset	Module Ablation	OA (%)	AA (%)	Ka	SDUA
Dataset-1	Basic	85.43	82.63	0.812	15.71
	Basic+TEA	89.98 ↑ 4.55	88.5 ↑ 5.87	0.869 ↑ 0.057	9.42 ↓ 6.29
	Basic+TEA+PLDC	91.42 ↑ 1.44	90.74 ↑ 2.24	0.888 ↑ 0.019	7.77 ↓ 1.65
	Basic+TEA+PLDC+FPS	**93.17** ↑ 1.75	**91.71** ↑ 0.97	**0.910** ↑ 0.022	**5.30** ↓ 2.47
Dataset-2	Basic	87.71	90.23	0.862	11.62
	Basic+TEA	91.78 ↑ 4.07	95.55 ↑ 5.32	0.908 ↑ 0.046	6.67 ↓ 4.95
	Basic+TEA+PLDC	92.45 ↑ 0.67	95.92 ↑ 0.37	0.915 ↑ 0.007	6.51 ↓ 0.16
	Basic+TEA+PLDC+FPS	**93.48** ↑ 1.03	**96.19** ↑ 0.27	**0.927** ↑ 0.012	**5.01** ↓ 1.40
Dataset-3	Basic	79.04	67.98	0.758	26.48
	Basic+TEA	83.40 ↑ 4.36	70.62 ↑ 2.64	0.807 ↑ 0.049	21.53 ↓ 4.95
	Basic+TEA+PLDC	84.04 ↑ 0.64	72.43 ↑ 1.81	0.813 ↑ 0.006	19.81 ↓ 1.72
	Basic+TEA+PLDC+FPS	**85.57** ↑ 1.53	**74.79** ↑ 2.36	**0.832** ↑ 0.009	**19.12** ↓ 0.69
Dataset-4	Basic	77.80	81.81	0.773	17.98
	Basic+TEA	83.95 ↑ 6.15	87.04 ↑ 5.23	0.826 ↑ 0.053	11.05 ↓ 6.93
	Basic+TEA+PLDC	86.43 ↑ 2.45	88.07 ↑ 1.03	0.853 ↑ 0.027	9.56 ↓ 1.49
	Basic+TEA+PLDC+FPS	**88.22** ↑ 1.79	**90.13** ↑ 2.06	**0.872** ↑ 0.019	**9.01** ↓ 0.55

IV. DISCUSSION

1) The application of the proposed SATE involves two parameters: parameter α in the increasing convex mapping function and sampling rate μ. In this section, the relationship between classification accuracy and the parameter setting of the proposed approach is discussed.

The parameter α critically influences sample quality and classification performance. The relationship between parameter α and the classification accuracy is investigated when μ is fixed to the optimal value (Fig. 8). The results indicate that the performance of the SATE optimally improves as α increases, and the classification performance gradually decreases with the continued increase of α. For instance, when $\alpha = 0.1$, the thresholds of all classes are consistently set at a high value, which may limit the augmentation of the training sample set. The OAs increase by 1.43% and 1.70% as α gradually increases from 0.1 to 0.3 for Dataset-1 and Dataset-2, respectively, reaching the optimal level. This improvement benefits from adaptive threshold adjustment to expand the initial training sample set. The OAs of Dataset-1 and Dataset-2 decrease from 93.17% and 93.48% to 85.78% and 90.10% with the continued increase of α from 0.3 to 0.9, respectively. Noise samples are introduced during the expansion of the initial sample set because a larger α will cause the lower limit of the threshold to decrease considerably. We set $\alpha = 0.2$ for Dataset-1 and Dataset-2 and $\alpha = 0.3$ for Dataset-3 and Dataset-4 in all the following experiments to achieve satisfactory classification results.

Unlike parameter α, parameter μ determines the size of the training sample set. The relationship between parameter μ and SATE classification performance is studied when α is fixed to the optimal value (Fig. 9). Different datasets exhibit varying fluctuations with the increment of μ in terms of the diverse evaluation metrics. For example, when μ increases from 0.1 to 0.4 for Dataset-1 and Dataset-4, OAs increase from 89.98% and 84.35% to 93.17% and 88.22%, respectively, and gradually maintain horizontal fluctuations with the continued increase of μ. The AA, Ka, and F1-score maintain similar floating changes as OA. A small μ will hinder the improvement of the expression of the distribution characteristics of ground objects. Meanwhile, a large μ means that a considerable number of samples is input to the classifier, which may result in sample redundancy. Accordingly, an optimal μ should be obtained by trial-and-error experiments when applying our approach in practical applications, thereby

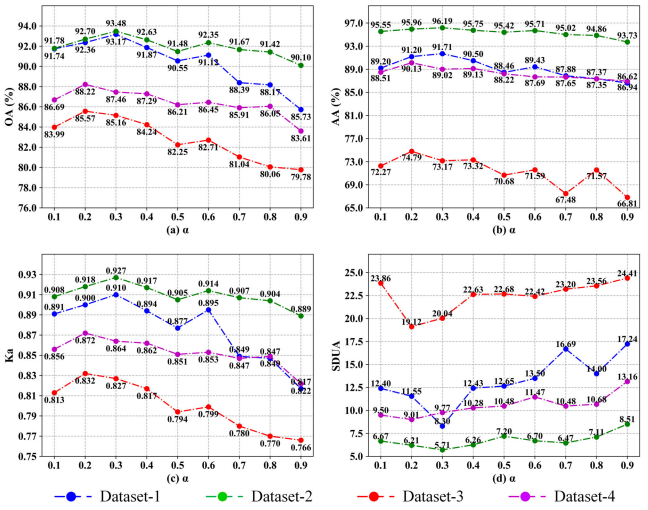


Fig. 8. Relationship between the classification accuracies and α of the proposed approach. (a) Relationship between OA and parameter α of the proposed approach. (b) Relationship between AA and parameter α of the proposed approach. (c) Relationship between Ka and parameter α of the proposed approach. (d) Relationship between SDUA and parameter α of the proposed approach.

effectively improving the classification performance of the SATE. Here, we set $\mu = 0.3$ for Dataset-2 and Dataset-3 and $\mu = 0.4$ for Dataset-1 and Dataset-4 to obtain the classification maps. Different datasets mean the different image scenes which are composed of different ground targets. The separate abilities among these targets are different for each dataset and the task of classification for these targets also requires different samples. Therefore, the optimized settings of α and μ are different for the classification of different targets. In the experiments, since the number of classes and coverage of HRSI data are significantly different from each other, in practice, the optimal configuration of parameters α and μ needs to be determined by trial and error. However, it can be seen that determining the optimized parameter setting via trial-and-error experiments is time-consuming. To improve the practical usability of the proposed approach and deduce the relationship between classification accuracies and parameter setting (Figs. 9 and 10), $\mu \geq 0.4$ and $\alpha \leq 0.3$ maybe a suitable choice in practical applications.

Fig. 10 illustrates the effect of τ_0 on the classification performance. The parameter τ_0 determines the upper threshold value for selecting highly confident pseudo-labeled samples. In general, as τ_0 increases, the values of the four objective metrics (OA, AA, Ka, and SDUA) become higher. It should also be noted that there is a small decrease in OA, AA, Ka, and SDUA when $\tau_0 = 1.0$. When the upper value is too low, the quantity of pseudo-labels capable of exceeding this value becomes larger; meanwhile, the confidence level deteriorates sharply. When the upper is fixed at a high value, the confidence of pseudo-labels is guaranteed, but the number of pseudo-labels drastically decreases. Based on the above analysis, τ_0 should be neither too large nor too small. In this work, τ_0 is set to 0.9 to achieve good results.

2) To verify the learning ability of the SATE under small-sample scenarios. Fig. 11 investigates the relationship between the number of samples per class and the classification accuracy. The performance of the SATE improves with the increase

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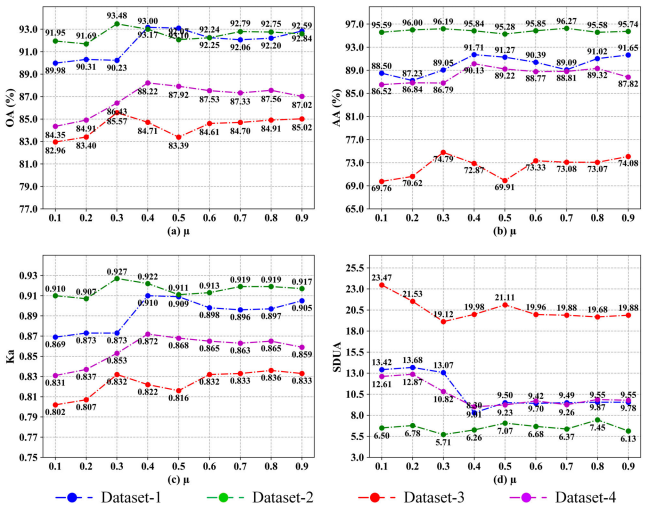


Fig. 9. Relationship between the classification accuracies and μ of the proposed approach. (a) Relationship between OA and parameter μ of the proposed approach. (b) Relationship between AA and parameter μ of the proposed approach. (c) Relationship between Ka and parameter μ of the proposed approach. (d) Relationship between SDUA and parameter μ of the proposed approach.

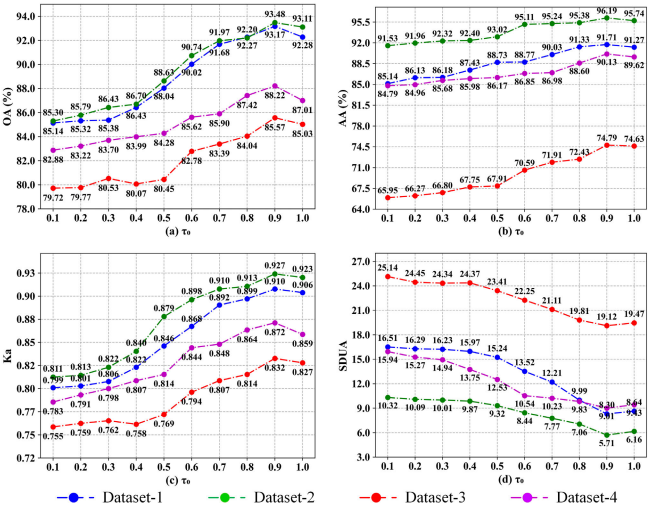


Fig. 10. Relationship between the classification accuracies and τ_0 of the proposed approach. (a) Relationship between OA and parameter τ_0 of the proposed approach. (b) Relationship between AA and parameter τ_0 of the proposed approach. (c) Relationship between Ka and parameter τ_0 of the proposed approach. (d) Relationship between SDUA and parameter τ_0 of the proposed approach.

in the number of initial samples per class. For example, the OA increases from 88.22% to 96.34% when the number of samples per class increases from 10 to 30. However, when the number of samples per class increases from 30 to 50, the improvement is small, such as the OA ranges from 96.34% to 97.42%. Similar to numerous classification methods, intuitively increasing the initial training samples per class improves the classification performance of the SATE, but it shows advantages when the initial samples are fewer than or equal to 30 samples per class. Moreover, the relationship between SDUA and the number of initial samples indicates that the balancing ability of the SATE can be improved with an increase in the number of initial samples. The various evaluation indicators exhibit diverse sensitivity levels to the different number of initial samples. Therefore, a tradeoff between the classification accuracies and

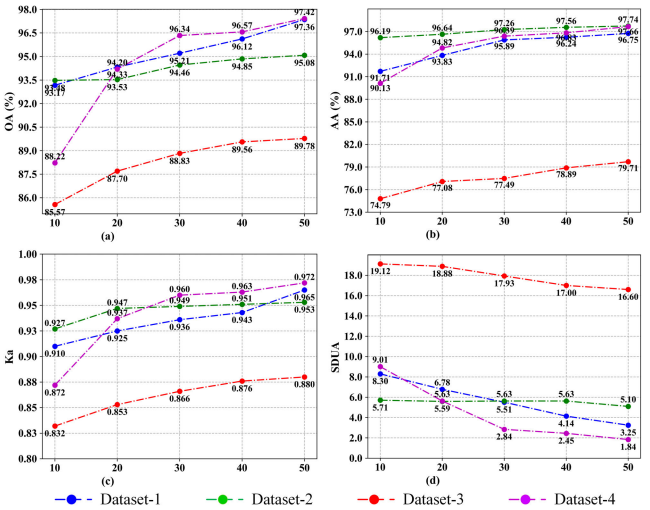


Fig. 11. Relationship between the classification accuracies and the number of initial training samples of the proposed approach. (a)–(d) Number of samples per class.

the labeling of initial samples is required when applying the SATE in practical applications.

3) The relationship between the total iteration of the SATE and the classification accuracy is discussed. The methodology section shows that the approach is iterative, and the SATE is terminated until all the classes satisfy a predefined condition. Because the balance between targets and the related samples affects the classification performance of a classifier, the samples are augmented dynamically in the progress of classification. In our proposed approach, when the user's accuracy for each class meets the predefined conditions [see (3)], the iteration will be terminated. In the experiments, different datasets contain different ground targets, so it means different iterations for our proposed approach. In other words, if a target refers to more pixels in an image scene, it needs more samples. Therefore, it needs more iteration for the sample augmentation. In Fig. 12, the OA, AA, and Ka increase with the increment of the iteration times from one to five. By contrast, SDUA decreases with the iteration from one to five. Thereafter, the OA, AA, Ka, and SDUA slightly change and tend to be at a horizontal level after the increment of iterations was larger than five. Fig. 12 also indicates that: 1) the SATE terminates at different iterations for various datasets, and the augmented samples using the SATE are effective in improving the classification accuracies with HRSIs and 2) the predefined iteration termination conditions are effective in balancing the user's accuracy for different datasets.

4) In addition, the time consumption of the proposed SATE and the deep learning methods are evaluated for each dataset, as shown in Table VIII. The comparison is conducted on a computer with Intel[1]Core[2] i7-7700-3.60 GHz CPU and 2080Ti GPU. The comparison indicated that the proposed SATE has better time complexity than 3D-HyperGAMO [50], ADGAN [29], and DM-MRN [39]. Compared with other methods, the proposed SATE needed more time to achieve the classification map for the same dataset, because the SATE was an iterative

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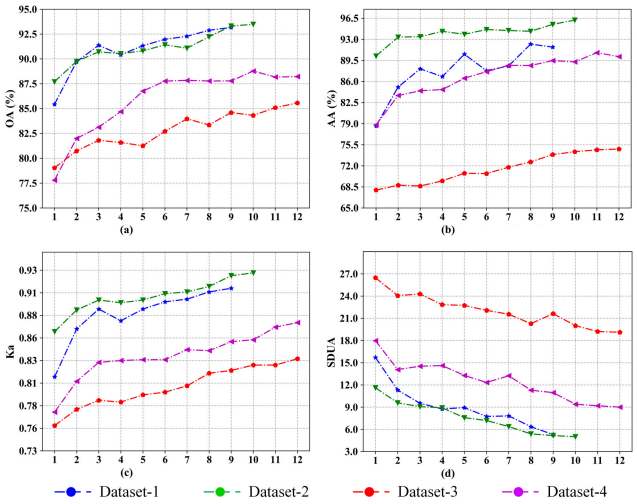


Fig. 12. Relationship between the classification accuracies and the iterations of the proposed approach. (a)–(d) Number of iterations.

TABLE VIII
TIME COST ON DIFFERENT DATASETS (THE UNIT IS MINUTE)

Methods	Dataset-1	Dataset-2	Dataset-3	Dataset-4
3D-GAN [28]	94.2	88.3	147.9	253.3
3D-HyperGAMO [50]	137.9	142.6	253.4	411.5
ADGAN [29]	164.3	168.4	286.3	446.8
3D-CNN [59]	37.5	32.5	75.4	134.2
DFSL [33]	123.6	114.55	257.9	396.4
S3-Net [34]	69.6	66.2	146.3	238.4
MDL-Net [36]	82.7	77.6	171.7	262.9
DM-MRN [39]	192.7	206.9	397.3	507.1
Proposed	135.9	145.2	247.5	397.6

algorithm that needed more time to augment and refine each class of samples.

V. CONCLUSION

This study proposes the SATE framework to address the dual challenges of limited samples and interclass imbalance in HRSI classification. By integrating adaptive threshold estimation (TEA), spatial pseudo-label correction (PLDC), and spectral diversity optimization (FPS), the SATE iteratively enhances both the quantity and quality of training samples, making it particularly suitable for scenarios with scarce annotations. The experimental results on four benchmark HRSI datasets indicated the feasibility of the proposed approach, and the comparisons with eight state-of-the-art methods demonstrated the superiority of the SATE. Although the application and comparisons with some real datasets efficiently verify the feasibility and advantages of the proposed approach, the SATE has the following limitations: 1) mislabeling may occur during the initial labeling process and 2) the SATE involves two parameters, and the optimal setting of the parameters is time-consuming for specific datasets. Therefore, in our future research, we will concentrate on developing noise–sample detection and correction methods to enhance the practical ability in applications. In addition, we will concentrate on developing a parameter estimation approach to improve the practical usability of the proposed approach.

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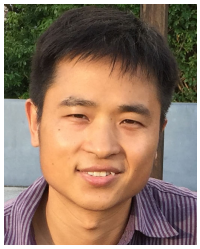
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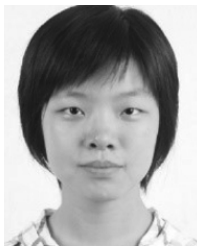
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